

deep metric learning:

image retrieval with deep networks

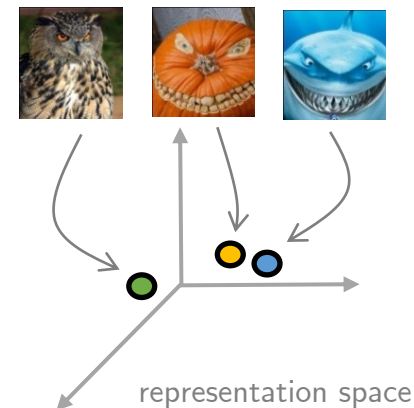
Computer Vision Methods 2024

CTU in Prague

Giorgos Tolias

metric learning

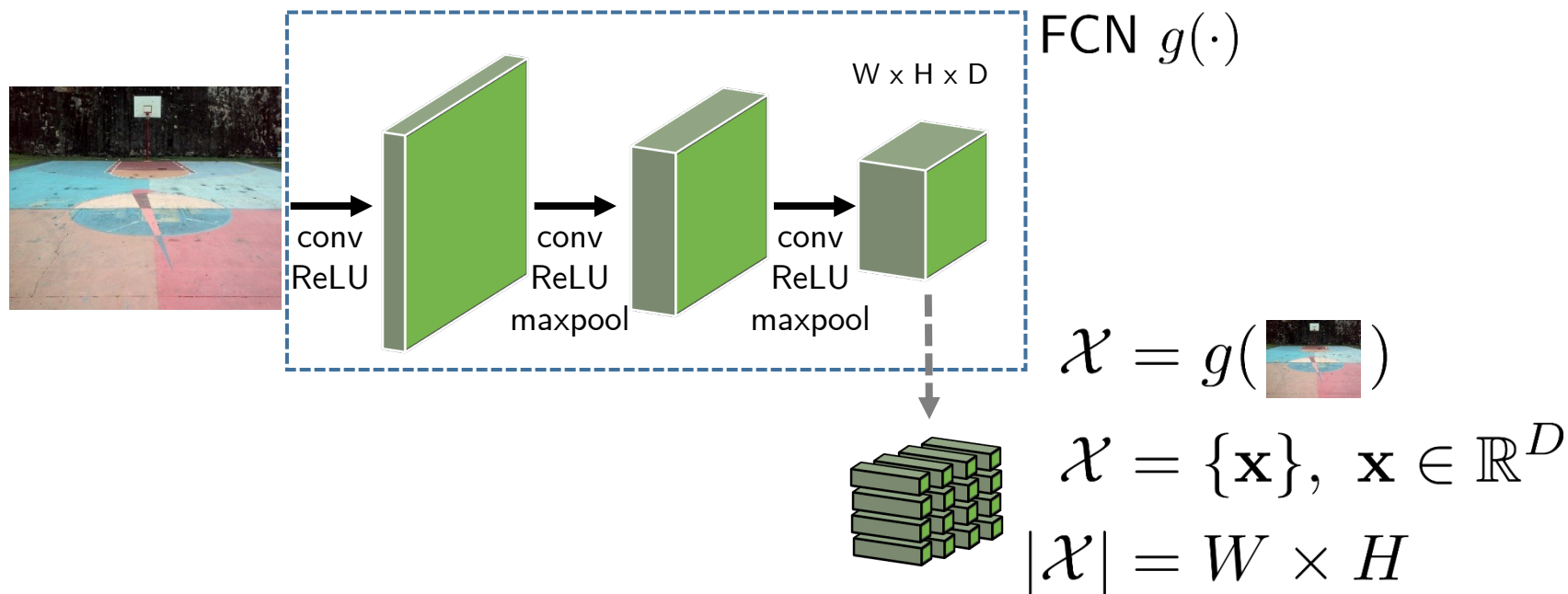
- learn metric $d(u, v) \in \mathbb{R}$, so that
 - relevant examples have small distances
 - non-relevant examples have high distances
- instead of directly learning $d(u, v)$
 - learn a representation function $f(u) \in \mathbb{R}^d$
 - for a standard metric in $\mathbb{R}^d$
- function f is a deep neural network
- deep learning for metric learning / retrieval
 - architectures for f
 - loss functions
 - training data



architectures

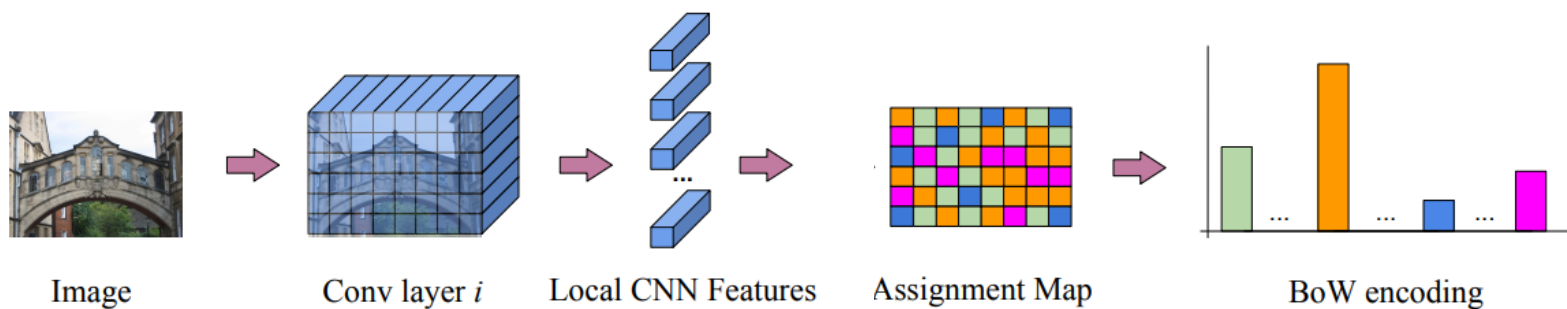
fully convolutional network

- fully convolutional network (FCN)
 - map images to 3D activation tensor with dimensionality $W \times H \times D$
 - no restrictions on input image size or aspect ratio
- activation tensor = densely extracted set of local descriptors
 - replaces classical local descriptors
 - apply classical approaches on top of deep local descriptors
 - end-to-end training: components should be trainable
 - transfer learning: FCN is pre-trained on a different task



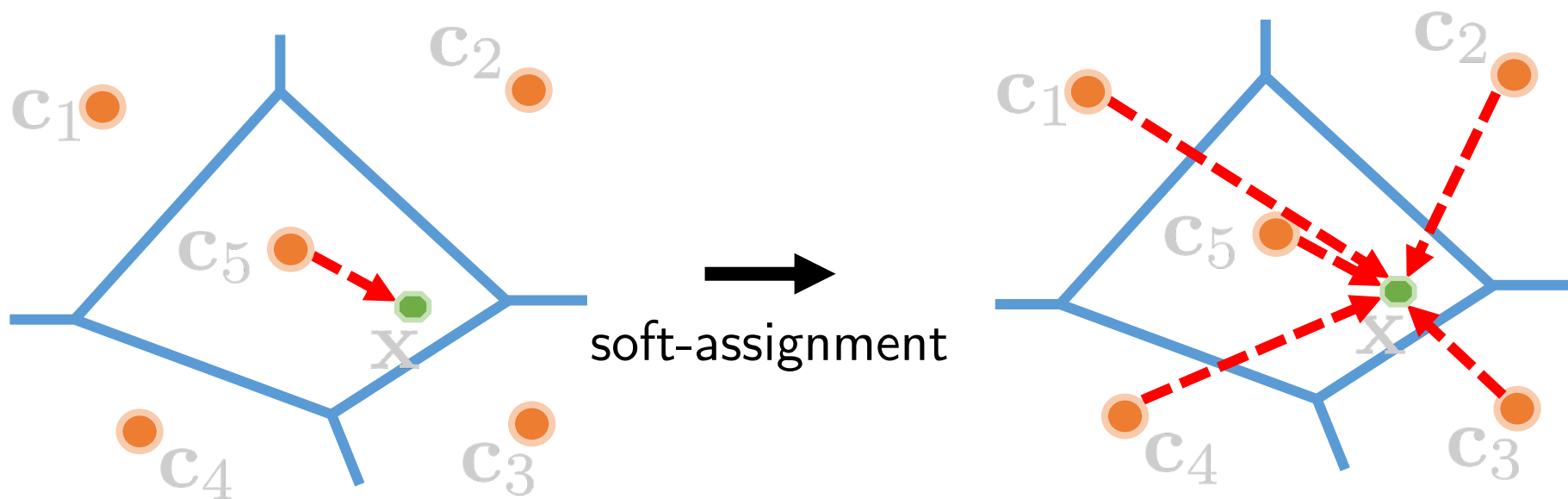
BoW with CNN features

- dense deep descriptors replace classical ones, eg SIFT
- apply BoW as with SIFT



NetVLAD

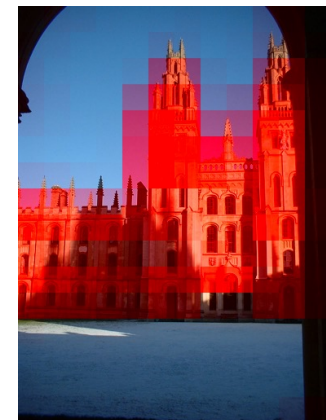
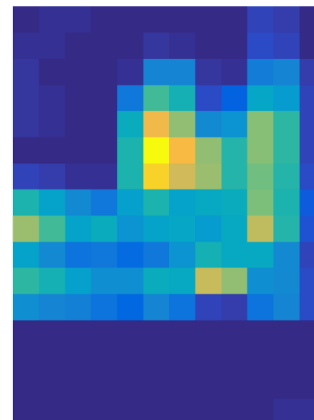
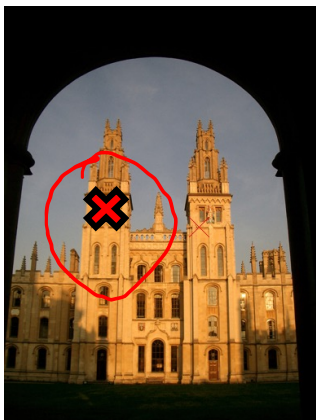
- VLAD on dense deep local descriptors
- the codebook is a learnable parameter matrix
- soft-assignment to all visual words
 - assignment weights: function of the distance to the centroid
 - more useful gradients in training



$$\phi(\mathbf{x}) = [w_1(\mathbf{x} - \mathbf{c}_1)^\top, \dots, w_i(\mathbf{x} - \mathbf{c}_i)^\top, \dots, w_K(\mathbf{x} - \mathbf{c}_K)^\top]^\top, \quad \sum_{i=1}^K w_i = 1$$

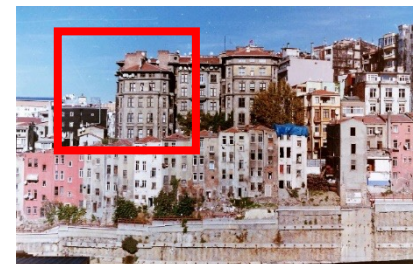
sum pooling – SPoC descriptor

- global sum pooling: $W \times H \times D \rightarrow 1 \times 1 \times D$
- global descriptor with sum pooling: $X \propto \sum_{\mathbf{x} \in \mathcal{X}} \mathbf{x}$
- image similarity: $X^\top Y \propto \left(\sum_{\mathbf{x} \in \mathcal{X}} \mathbf{x} \right)^\top \sum_{\mathbf{y} \in \mathcal{Y}} \mathbf{y} = \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{y} \in \mathcal{Y}} \mathbf{x}^\top \mathbf{y}$
- simple but works well: discriminative power of CNN activations
- translation invariant representation



max pooling – MAC descriptor

- global max pooling: $W \times H \times D \rightarrow 1 \times 1 \times D$
- activation tensor as set of D feature maps ($W \times H$ matrices)
- keep maximum value per feature map (channel)
- each descriptor element corresponds to a feature map
 - implicitly corresponds to an image region



input image



conv₅ - channel 1



conv₅ - channel 2

....



conv₅ channel i

....



conv₅ - channel D

maximum activation

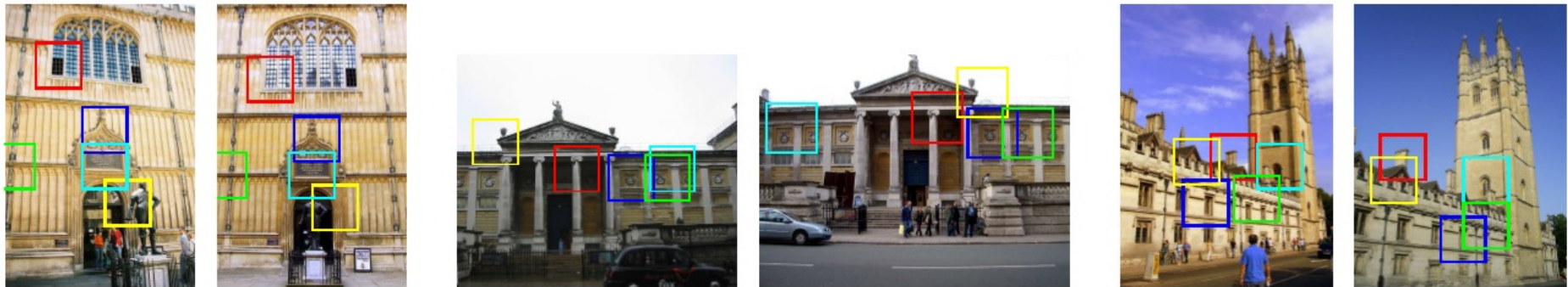


- representation: $X = [\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(i)}, \dots, \mathbf{x}_{(D)}] \in \mathbb{R}^D$

[Razavian et al., MTA'16]
[Tolias et al., ICLR'16]

max pooling – MAC descriptor

- image-to-image similarity = MAC descriptor dot product
- implicitly takes region correspondences into account
 - each descriptor element corresponds to an image region
 - see which elements contribute most to the dot product

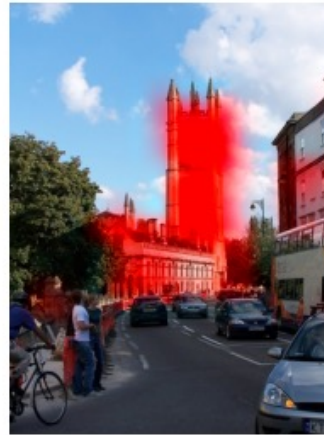
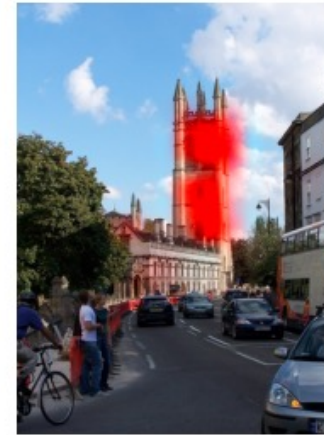
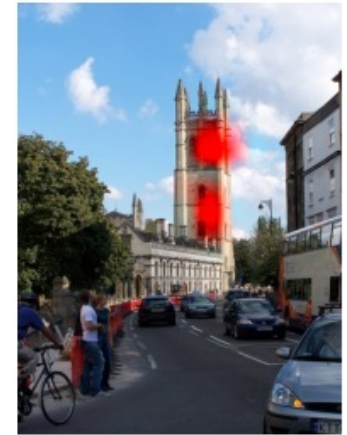


receptive field for activations that contribute the most to MAC descriptors dot product

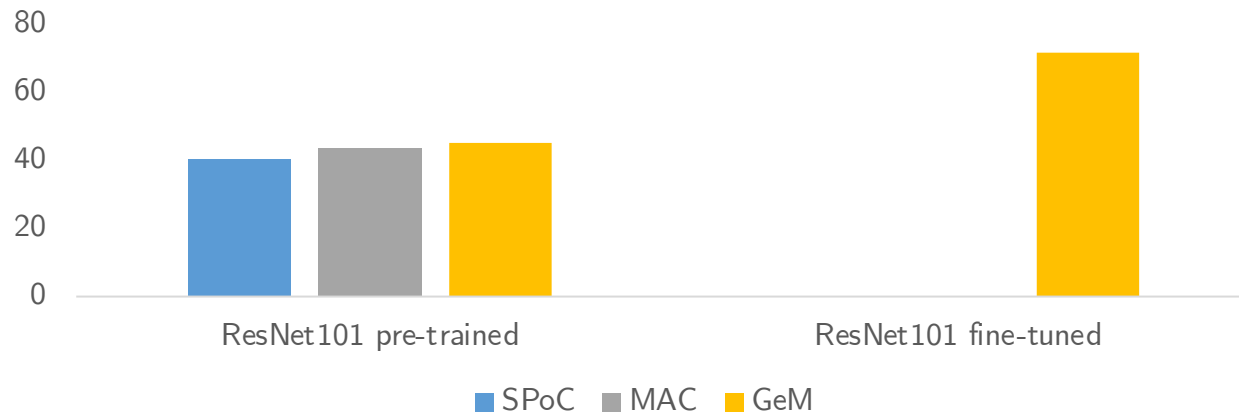
generalized mean pooling – GeM descriptor

- generalized mean pooling
 - generalizes average and max pooling
 - $p \rightarrow \infty$: max pooling (MAC)
 - $p = 1$: average pooling (SPoC)
- parameter p can be fixed or learned

$$X \propto \left(\frac{1}{|\mathcal{X}|} \sum_{\mathbf{x} \in \mathcal{X}} \mathbf{x}^p \right)^{\frac{1}{p}}$$

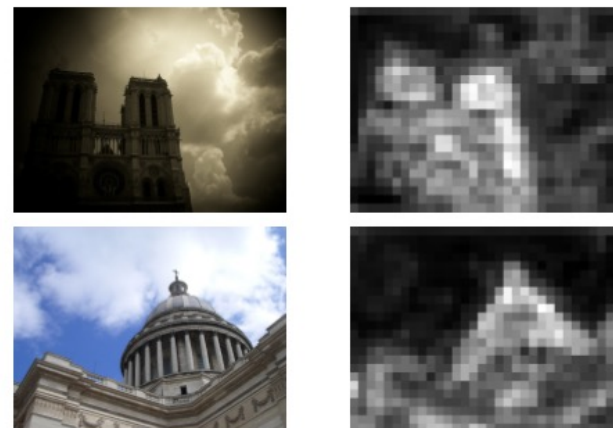
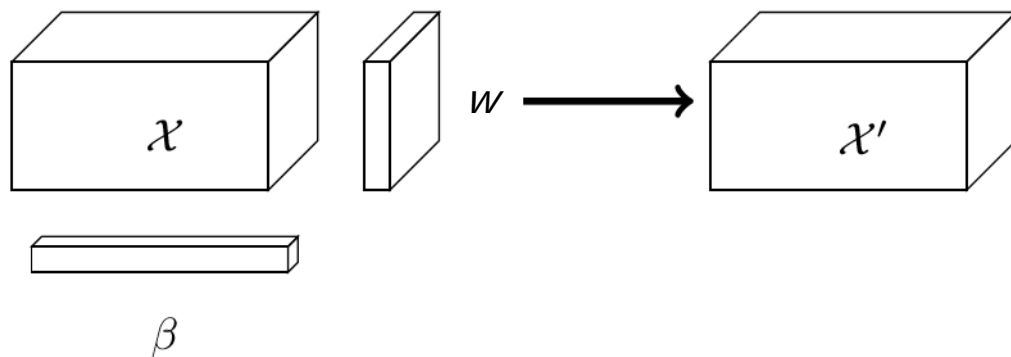
 $p = 1$  $p = 3$  $p = 10$

performance results



mAP on R-Oxford with database of 1m images

weighted sum pooling – CroW descriptor



example of w

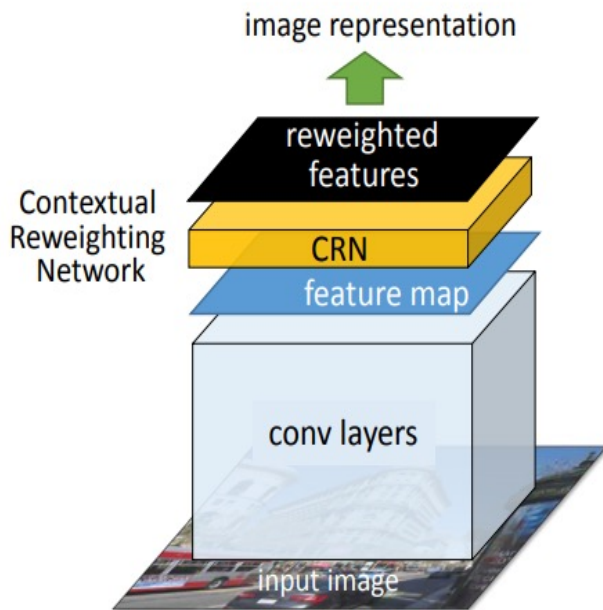
w : attention weight based on L2 norm of local descriptors
 β : inverted-document-frequency weight

- closer look at attention weights w

$$X \propto \sum_{\mathbf{x} \in \mathcal{X}} w_x \mathbf{x} \quad \leftarrow \text{global descriptor via weighted aggregation}$$

$$X^\top Y \propto \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{y} \in \mathcal{Y}} w_x w_y \mathbf{x}^\top \mathbf{y} \quad \leftarrow \text{similarity via weighted vote accumulation}$$

weighted sum pooling – attention map



$$X \propto \sum_{\mathbf{x} \in \mathcal{X}} w_x \phi(\mathbf{x})$$

$$X^\top Y \propto \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{y} \in \mathcal{Y}} w_x w_y \phi(\mathbf{x})^\top \phi(\mathbf{y})$$

- attention weights estimated in a learnable way
 - small network on top of the 3D activation tensor

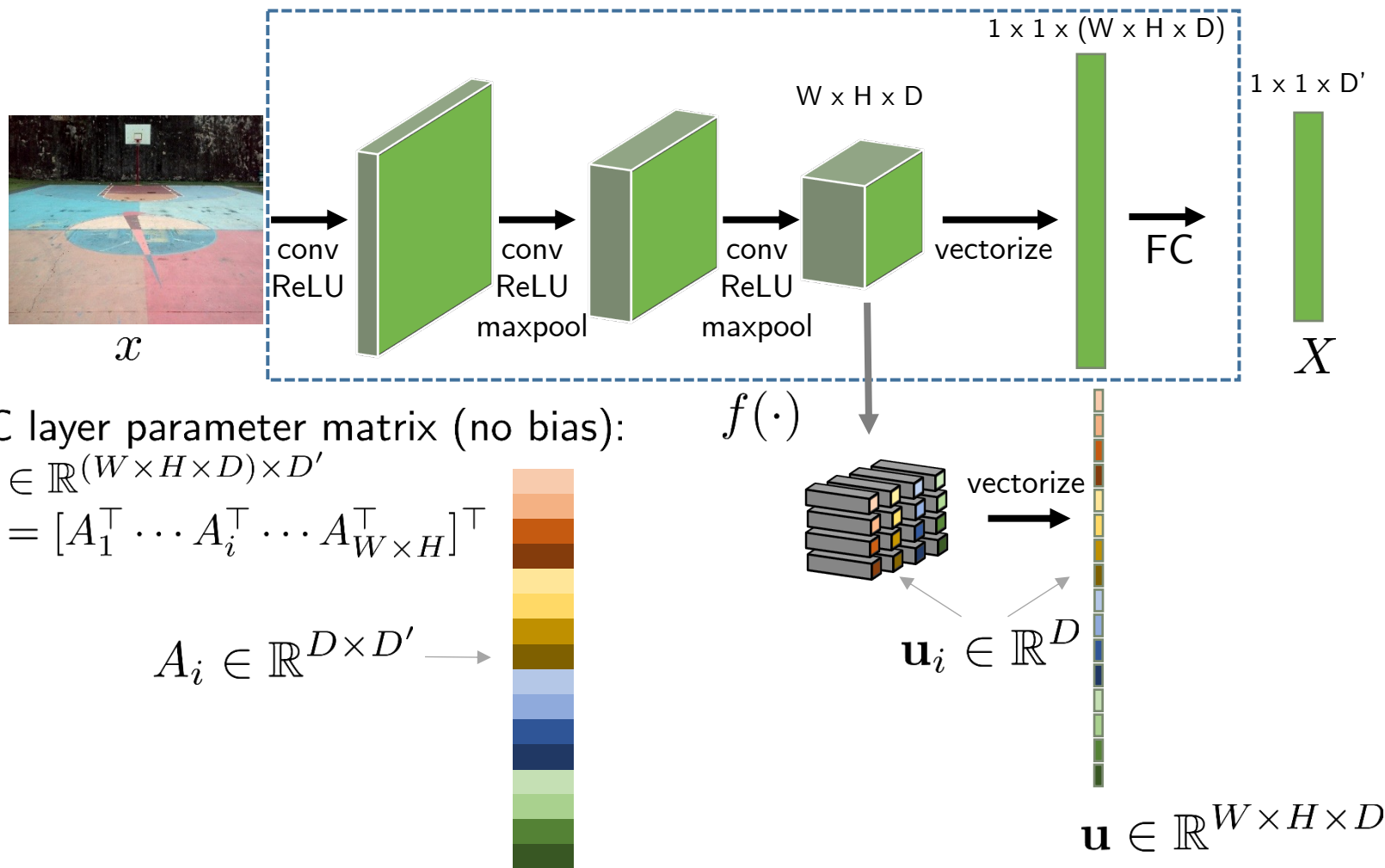
NetVLAD accuracy, San Francisco

learned attention: **83.2** vs CroW attention : **80.1**

NetVLAD accuracy, Tokyo 24/7

learned attention : **75.2** vs no attention : **71.8**

flatten+FC layer representation



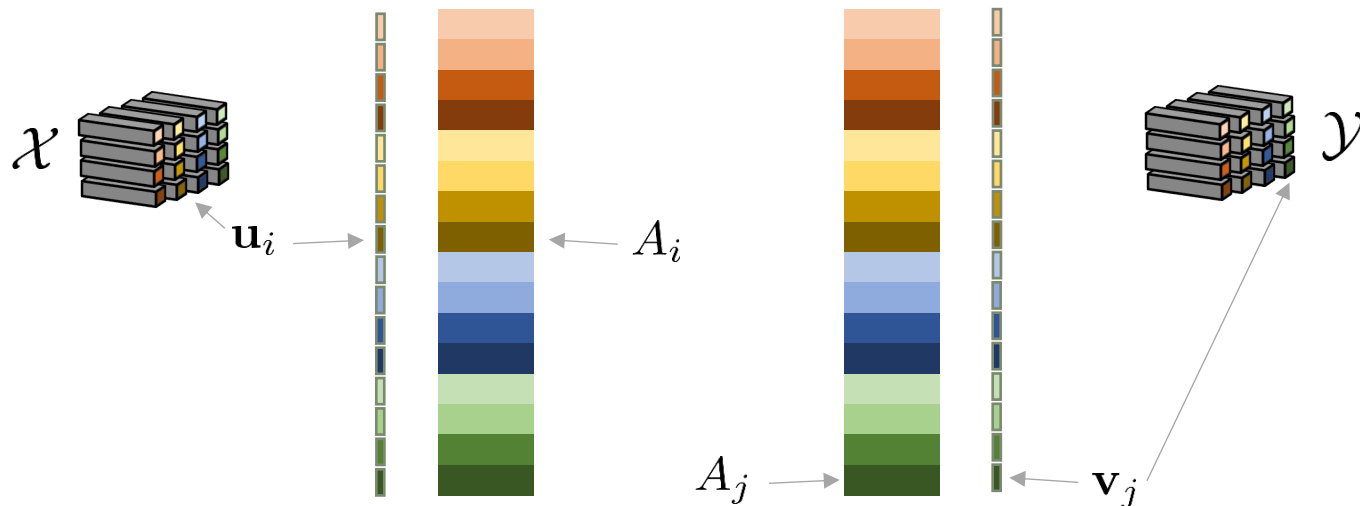
- representation: $X \propto A^\top \mathbf{u} = \sum_{i=1 \dots W \times H} A_i^\top \mathbf{u}_i \in \mathbb{R}^{D'}$

flatten+FC layer representation

- image-to-image similarity:

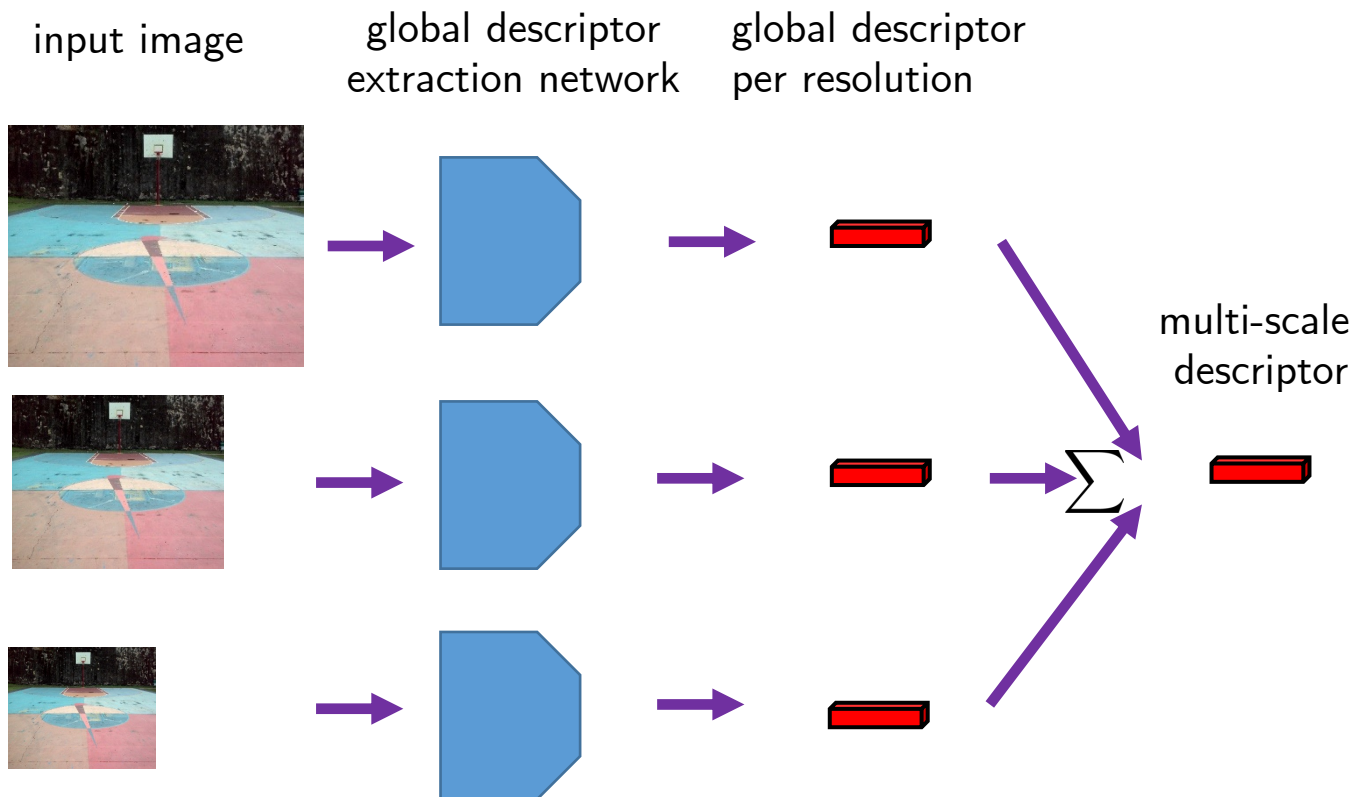
$$X^{\top}Y = \left(\sum_{i=1 \dots W \times H} A_i^{\top} \mathbf{u}_i \right)^{\top} \sum_{j=1 \dots W \times H} A_j^{\top} \mathbf{v}_j = \sum_{i=1 \dots W \times H} \sum_{j=1 \dots W \times H} \mathbf{u}_i^{\top} A_i A_j^{\top} \mathbf{v}_j$$

- translation variant representation

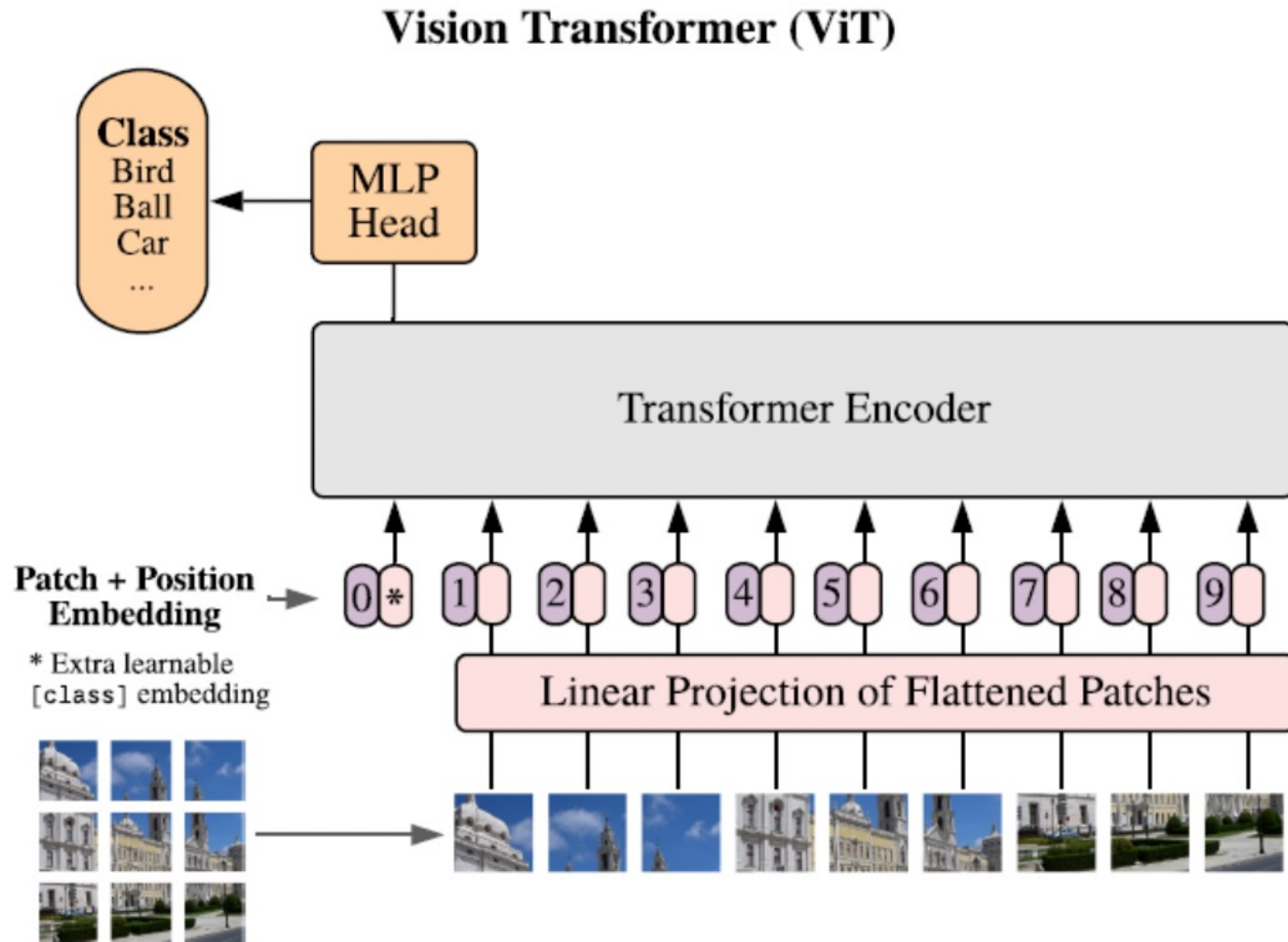


multi-scale representation

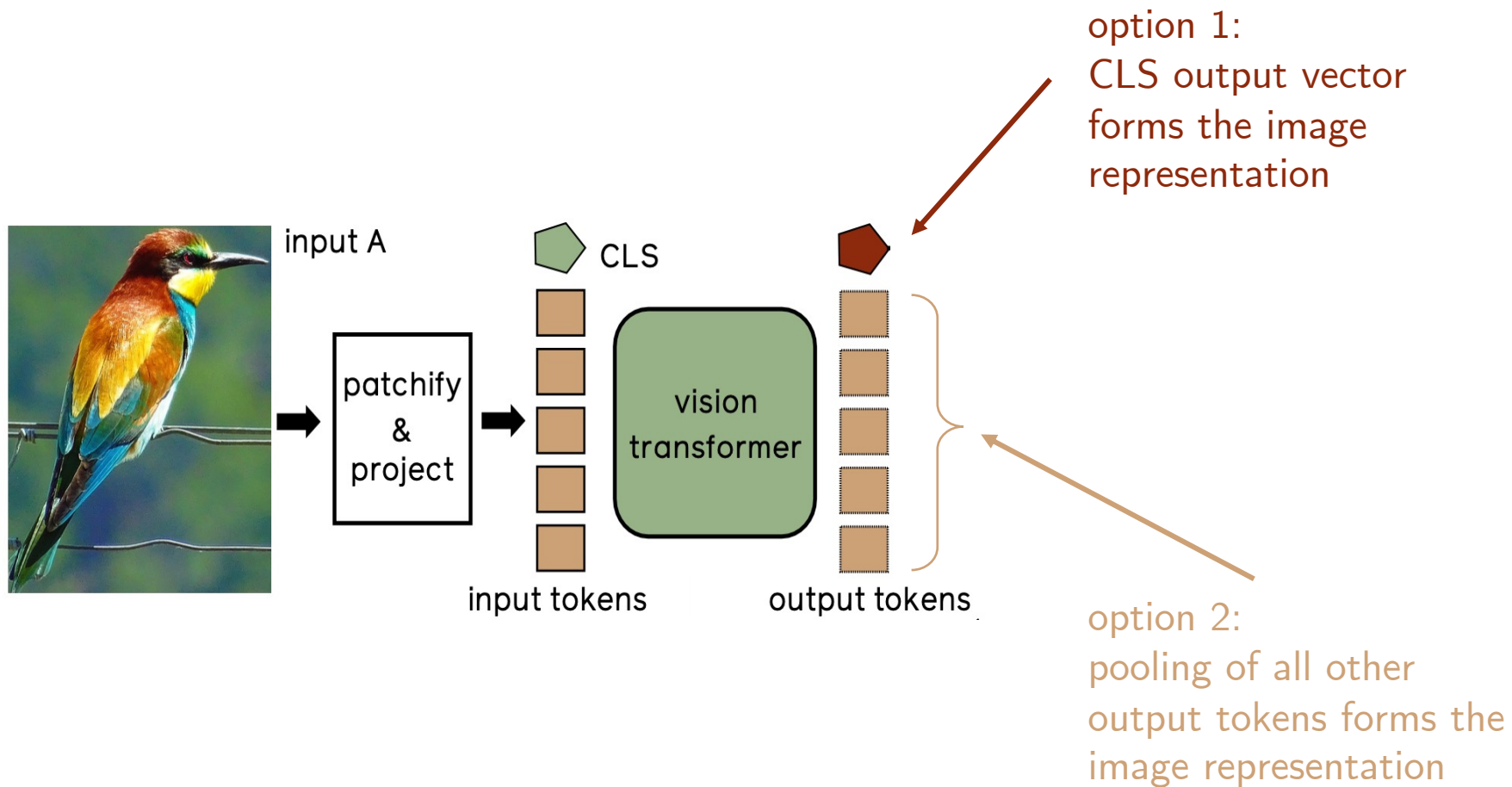
- small tolerance to scale-change
- create multi-scale representation
 - global descriptor for multiple resolutions
 - sum-pooling of global descriptors



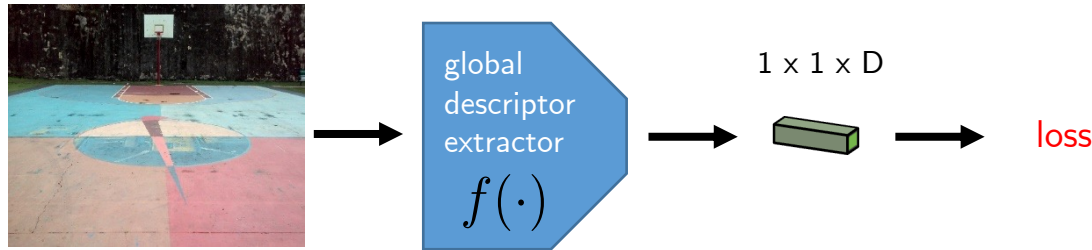
representation with ViT



representation with ViT



network to extract global descriptors



■ labels

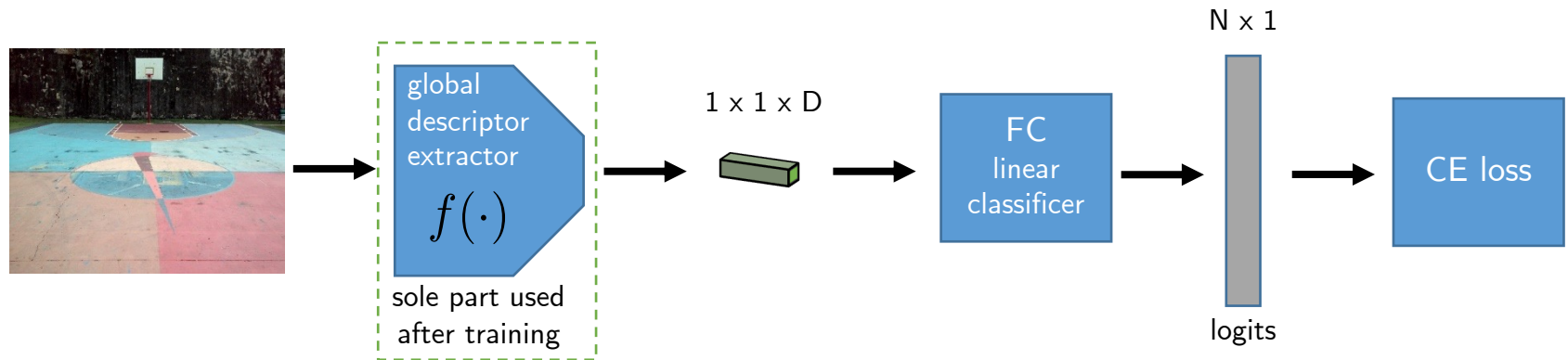
- discrete class labels: each image assigned to a category
- pairwise labels: two images are relevant or non-relevant

■ loss

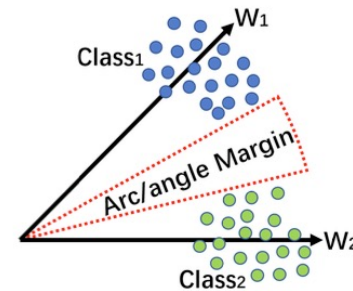
- classification losses
- pairwise losses

losses

classification loss



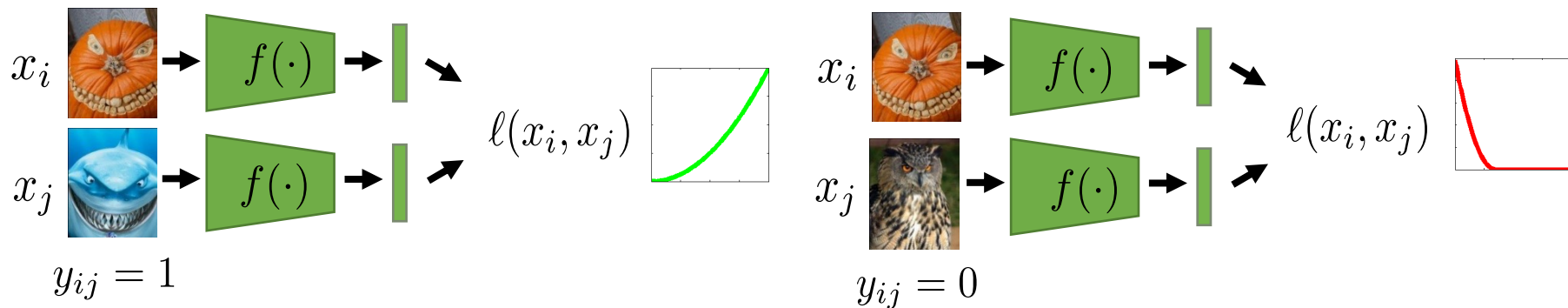
- train an N-way classifier
 - categorical supervision: class labels
 - classification loss
 - cross-entropy: standard choice
 - arc-face loss [Deng et al, CVPR'19]
 - sub-center arc-face [Deng et al, ECCV'20]



- network repurposing after training
 - discard the classifier
 - use descriptor extractor for retrieval
- not directly optimizing pairwise similarity
 - solution: use pairwise labels and pairwise loss

contrastive loss

- two-branch network; 2 networks that share weights



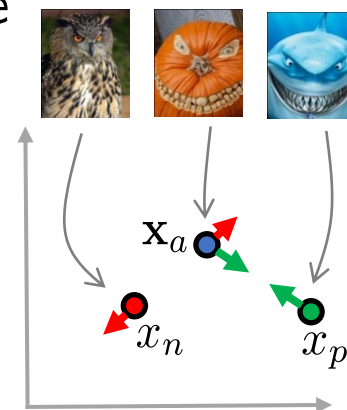
$$\ell(x_i, x_j) = \frac{1}{2}y_{ij}\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 + \frac{1}{2}(1 - y_{ij})[\tau - \|\mathbf{x}_i - \mathbf{x}_j\|_2]_+^2$$

- pushes descriptors of a positive pair as close as possible
- pushes descriptors of a negative pair far enough
 - “enough” defined by the margin hyper-parameter

$$\frac{\partial \ell}{\partial \mathbf{x}_a} = \mathbf{x}_a - \mathbf{x}_p \quad \frac{\partial \ell}{\partial \mathbf{x}_a} = \frac{\tau - \|\mathbf{x}_a - \mathbf{x}_n\|}{\|\mathbf{x}_a - \mathbf{x}_n\|} (\mathbf{x}_n - \mathbf{x}_a)$$

similar pair

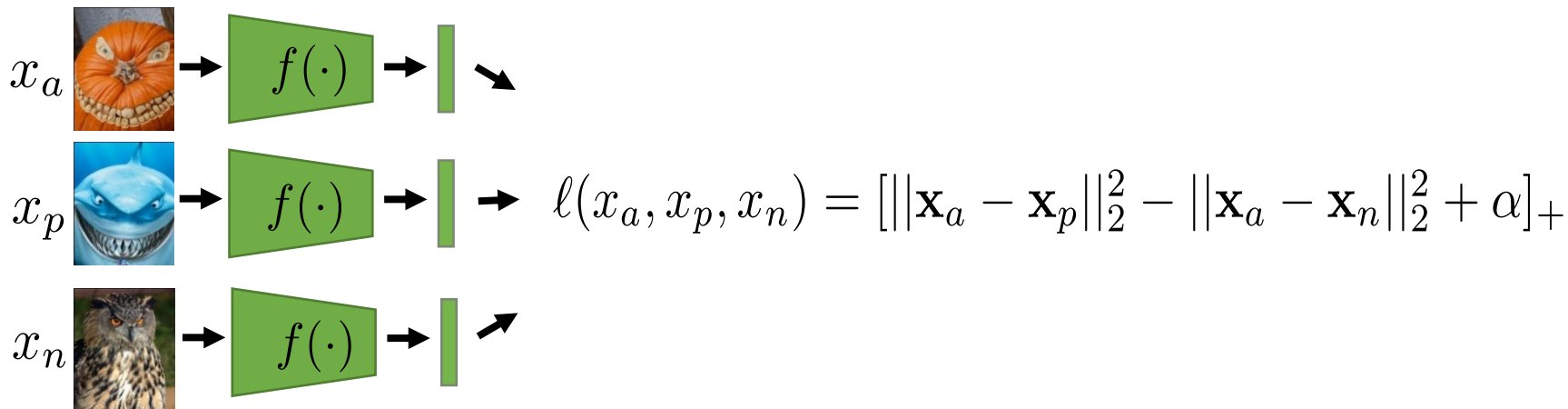
dissimilar pair



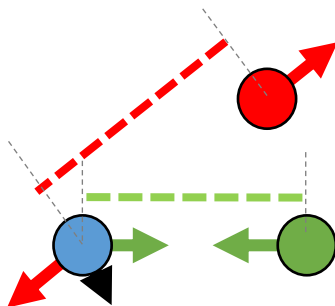
[Hadsell et al. CVPR'06]

triplet loss

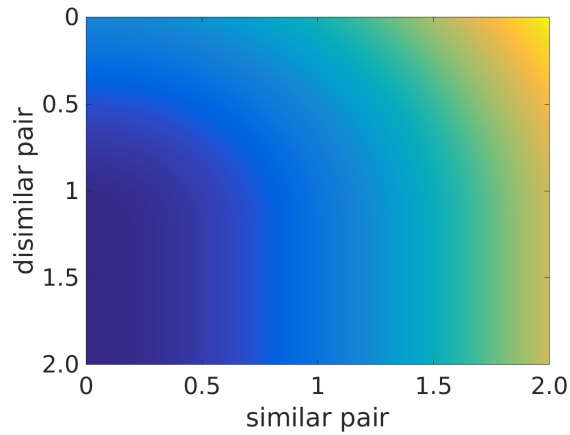
- three-branch network; 3 networks that share weights



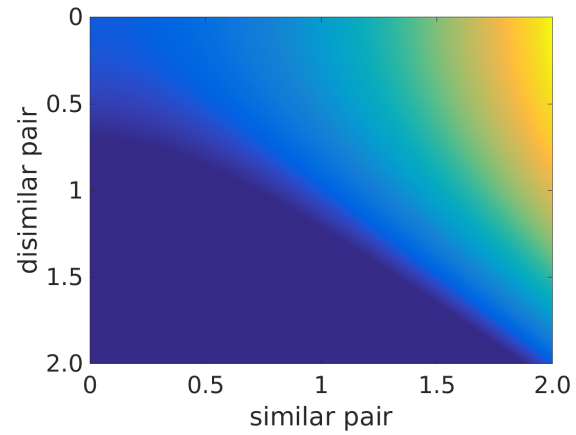
- triplet formed by one positive and one negative pair for the same anchor image
- non-zero gradients only if the positive pair is close enough compared to the negative pair



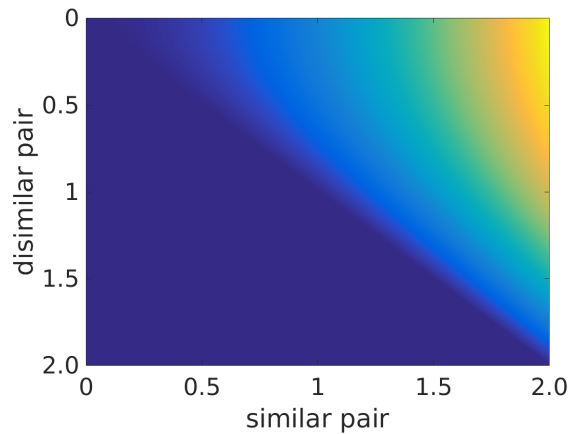
loss landscape



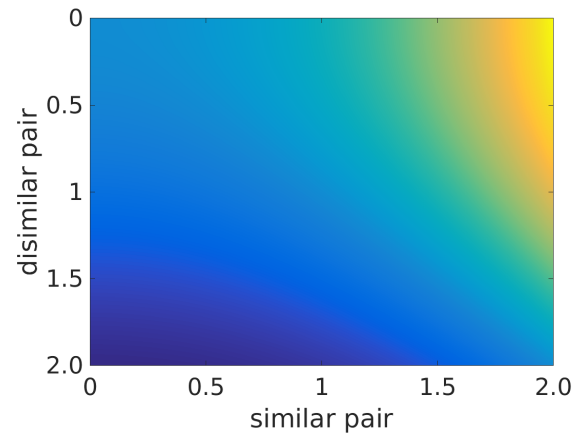
contrastive



triplet



$$[\mathbf{x}_a^\top \mathbf{x}_n - \mathbf{x}_a^\top \mathbf{x}_p]_+$$



$$\log(1 + e^{\mathbf{x}_a^\top \mathbf{x}_n - \mathbf{x}_a^\top \mathbf{x}_p}) \quad [\text{Sohn 2016}]$$

hard-negatives

- hard-negative mining is important
 - typically, most pairs are negative - cannot include all
 - many negatives already have far enough descriptor
 - zero gradients – nothing to learn
 - choose hard ones – nearby in the descriptor space
 - find nearest according to the current network parameters
- mining within the whole training set
 - hardness of each pair keeps changing
 - hard-negative mining repeated frequently during training
- mining within the batch
 - memory constraints keep the batch-size small
 - the larger the batch, the more likely to include hard negatives in the batch even with random sampling

histogram loss

[Ustinova & Lempitsky, 2016]

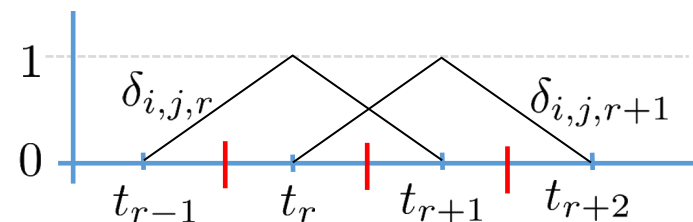
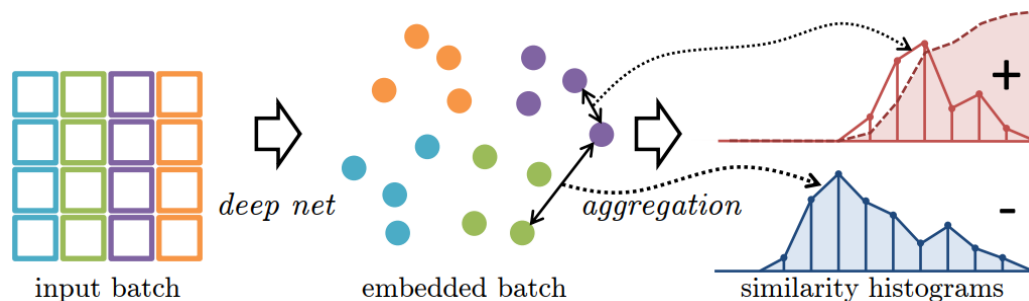
- minimize probability that similarity of a random negative pair is higher than the similarity of a random positive pair

- $$\mathbb{E}_{u \sim p^-} [\Phi^+(u)] = \int_{-1}^1 p^-(u) \Phi^+(u) \, du = \int_{-1}^1 p^-(u) \left[\int_{-1}^u p^+(v) \, dv \right] \, du$$

- approximated by
$$\sum_{r=1}^R \left(h_r^- \sum_{q=1}^r h_q^+ \right) = \sum_{r=1}^R (h_r^- \phi_q^+)$$

- histogram for positive pairs:
$$h_r^+ = \frac{1}{|\mathcal{P}^+|} \sum_{(i,j):(x_i,x_j) \in \mathcal{P}^+} \delta_{i,j,r}$$

- equivalently for the negative pairs



optimize the evaluation metric

- retrieval evaluation metrics are not differentiable
 - mean Average Precision (mAP)
 - recall @ k
 - precision @ k
- optimize a differentiable surrogate instead
 - AP loss [Revaud et al. ICCV 2019] [He et al, CVPR 2018]
 - Smooth AP [Brown et al. ECCV 2020]
 - Recall@k surrogate [Patel et al. arxiv 2021]

recall@k

query



database images



0.6



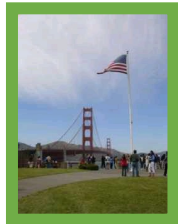
0.9



0.7



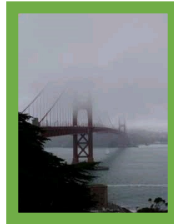
0.8



0.9



0.8



0.7



0.6

sort

number of positives with ranking less than k

$$\text{Recall @ 2} = \frac{1}{2}$$

recall@k

query



database images

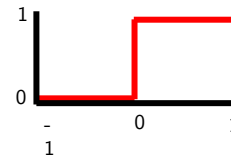


$$\text{ranking} = 1 + 2 \\ = 3$$

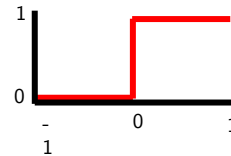
1

1

0



$$0.7 - 0.9 = -0.2$$



$$0.7 - 0.6 = 0.1$$

recall@k

query



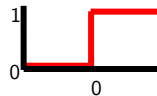
database images



0.6

4

$2-1-4=-3$



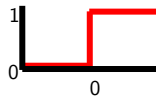
0



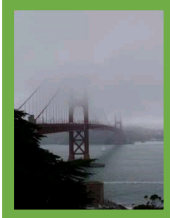
0.9

1

$2-1-1=0$



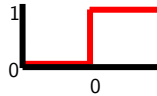
1



0.7

3

$2-1-3=-2$



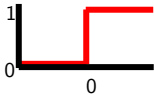
0



0.8

2

$2-1-2=-1$



0

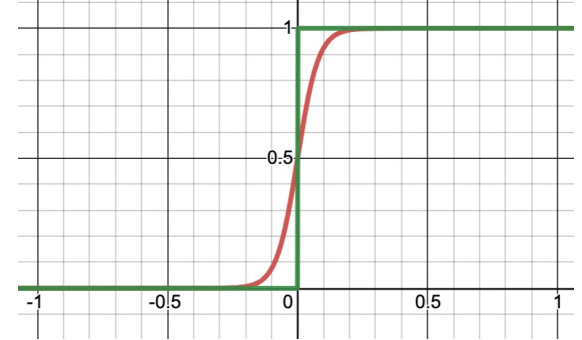
$k = 2$

Recall @ 2 = $\frac{1}{2}$

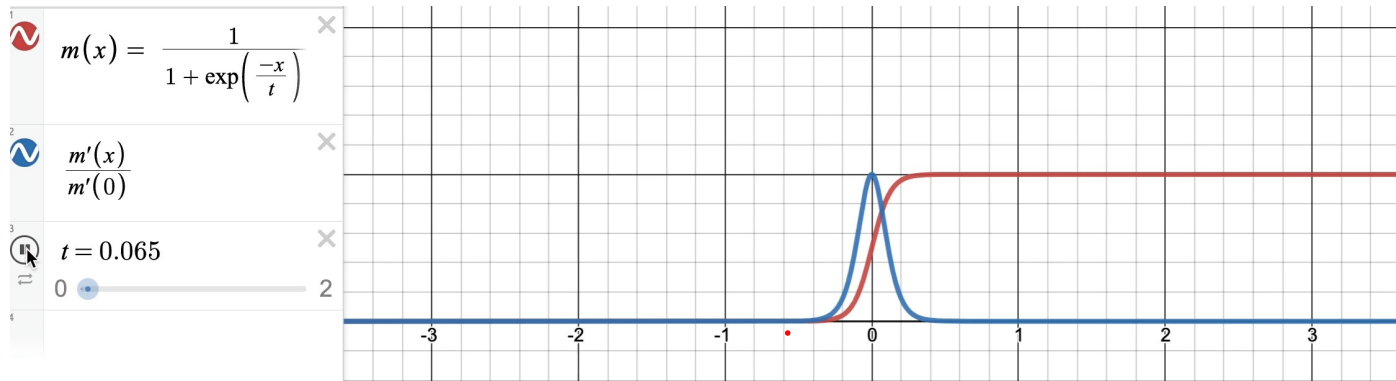


recall surrogate

- replace step function with appropriate sigmoid



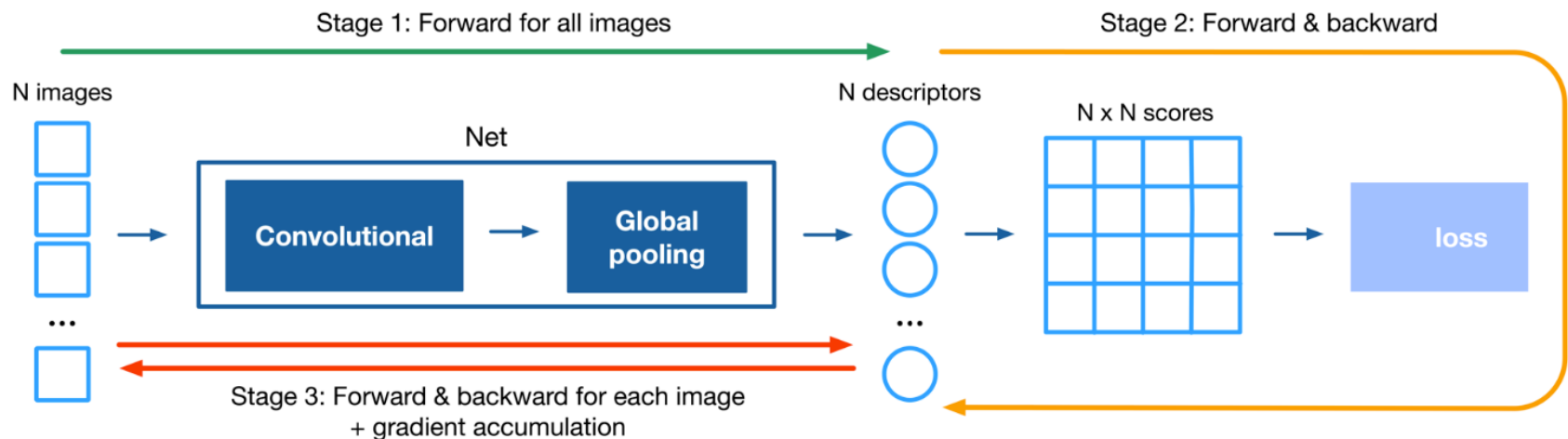
- trade-off: approximation vs non-zero gradient



- recall surrogate loss vs contrastive loss (GeM descriptor)
 - 46.1 vs 44.3 on Roxford-H
 - 63.9 vs 61.5 on Rparis-H

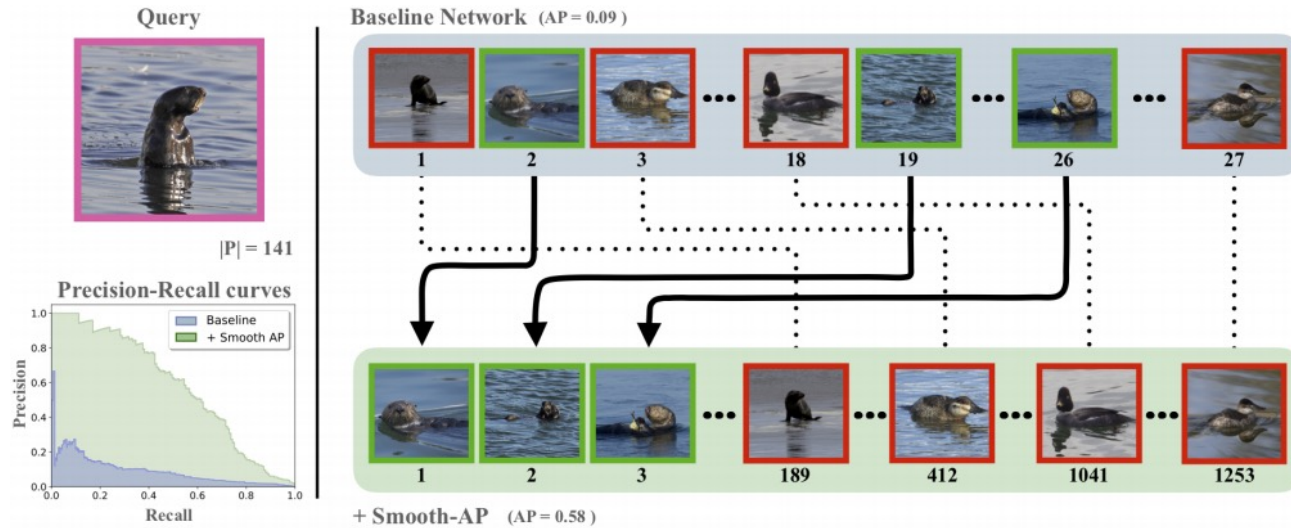
very large batch size

- large batch size is essential to optimize recall
- 3 step approach – implementation trick
 - extra computational cost
 - but significantly increases the maximum batch size



[Revaud et al., ICCV'19]

smooth AP loss



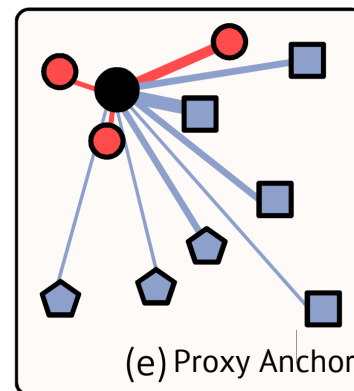
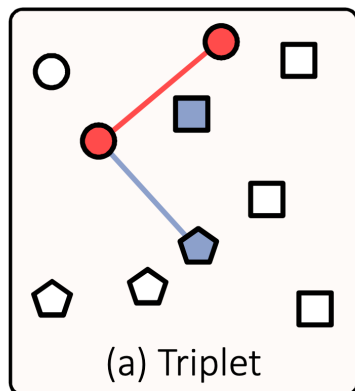
- Average-Precision (AP) is a common retrieval metric

$$\begin{aligned} \text{AP}_q &= \frac{1}{|\mathcal{S}_P|} \sum_{i \in \mathcal{S}_P} \text{precision@ranking}_i \\ &= \frac{1}{|\mathcal{S}_P|} \sum_{i \in \mathcal{S}_P} \frac{\#\text{positives-up-to-ranking}_i}{\text{ranking}_i} \\ &= \frac{1}{|\mathcal{S}_P|} \sum_{i \in \mathcal{S}_P} \frac{\mathcal{R}(i, \mathcal{S}_P)}{\mathcal{R}(i, \mathcal{S}_\Omega)} \end{aligned}$$

- AP is not differentiable
- optimize a smooth approximation instead [Brown et al. 2020]

proxy anchor

- proxy: a learnable vector per class
- optimize distances between training examples and proxies



$$\begin{aligned}
 \ell(X) = & \frac{1}{|P|} \sum_{p \in P} \log \left(1 + \sum_{x \in X_p^+} e^{-\alpha(s(x,p) - \delta)} \right) \\
 & + \frac{1}{|P|} \sum_{p \in P} \log \left(1 + \sum_{x \in X_p^-} e^{\alpha(s(x,p) + \delta)} \right)
 \end{aligned}$$

all proxies

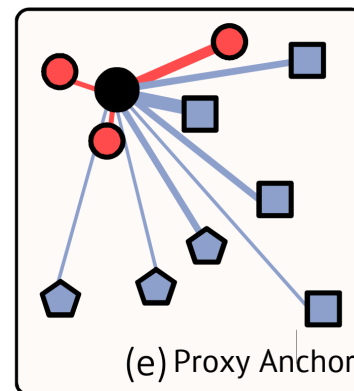
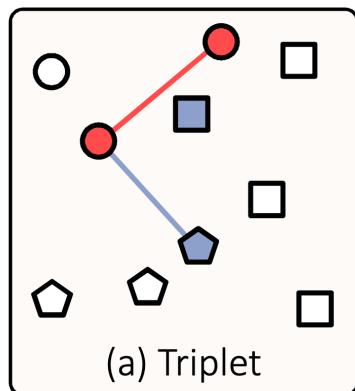
all training examples from this proxy/class

LogSumExp approx. max

all training examples NOT from this proxy/class

proxy anchor

- proxy: a learnable vector per class
- optimize distances between training examples and proxies



$$\frac{\partial \ell(X)}{\partial s(x, p)} = \begin{cases} \frac{1}{|P^+|} \frac{-\alpha h_p^+(x)}{1 + \sum_{x' \in X_p^+} h_p^+(x')}, & \forall x \in X_p^+, \\ \frac{1}{|P|} \frac{\alpha h_p^-(x)}{1 + \sum_{x' \in X_p^-} h_p^-(x')}, & \forall x \in X_p^-, \end{cases} \quad (6)$$

where $h_p^+(x) = e^{-\alpha(s(x, p) - \delta)}$ and $h_p^-(x) = e^{\alpha(s(x, p) + \delta)}$

training data

labels for image pairs

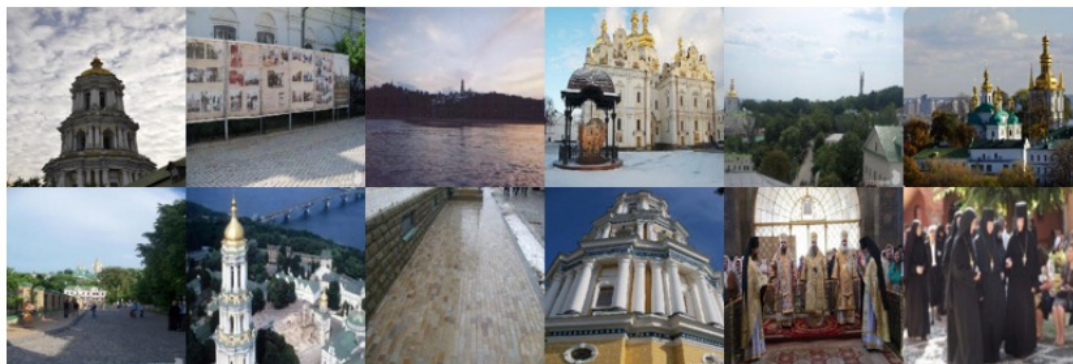
- need pairwise labels for ground-truth
 - matching pair, also called positive or relevant
 - non-matching pair, also called negative or non-relevant
- pairwise labels are costly to obtain
 - cost quadratic to the number of images in the training set
- solutions:
 - proxy labels
 - weak labels
 - automatically obtained labels
 - self-supervision: tasks where labels are available for free

class labels

- if discrete image labels are available (as in classification)
 - image pairs from different classes are labeled as non-matching
 - image pairs from the same class are labeled as matching
- not very useful for pairwise losses (and instance-level recognition)
 - matching pairs do not necessarily depict the same object view



Leeds
Castle



Kiev Pechersk
Lavra

weak labels - GPS

- use GPS information as weak labeling
- far enough images are negatives
- nearby images are possibly positives
 - camera orientation is unknown – might depict different objects
 - use descriptor distance to disambiguate – closest descriptor



● candidate negatives

● anchor



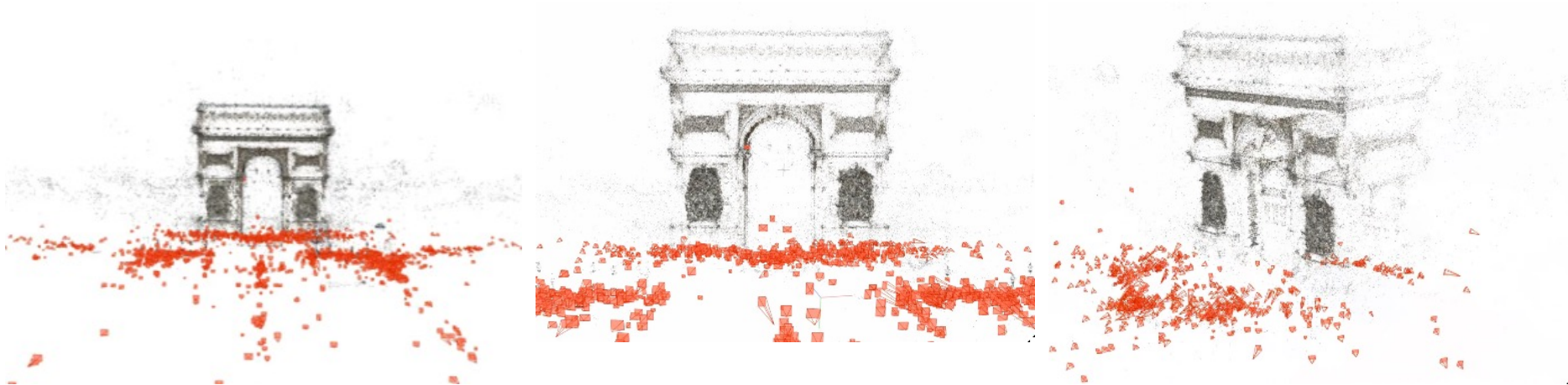
● candidate positives

▲ camera orientation (unknown)

[Arandjelovic et al., CVPR'16]

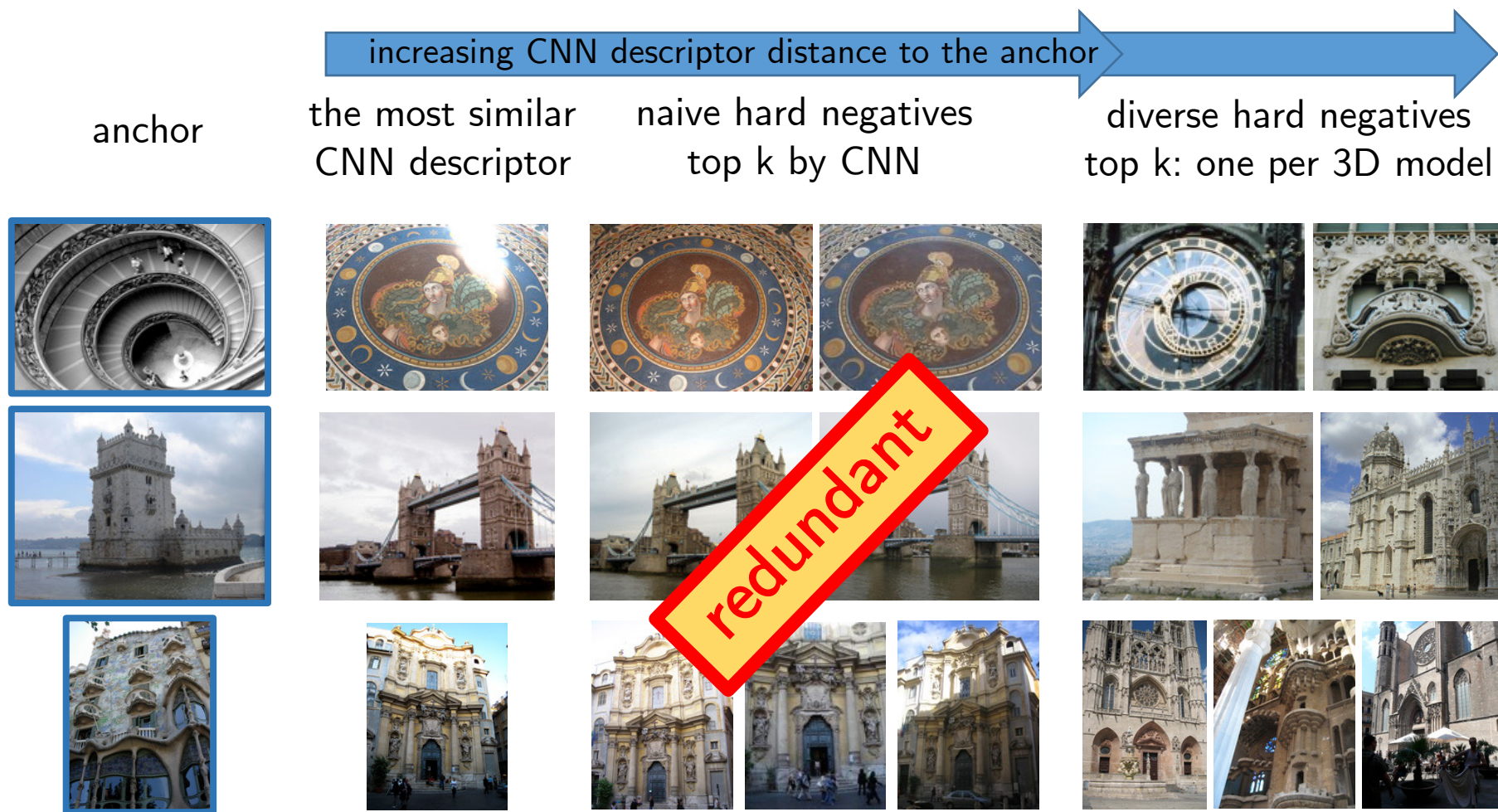
structure-from-motion - SfM

- use result of large-scale SfM to pick training pairs
 - camera orientation known
 - number of inliers known
- 7.4M images $\rightarrow$ 713 training 3D models



hard negatives from SfM

- negative examples: images from different 3D models than the anchor
- hard negatives: closest negative examples to the anchor



hard positives from SFM

- **positive examples:** images that share 3D points with the anchor
- **hard positives:** positive examples not close enough to the anchor

anchor

top 1 by CNN

top 1 by SfM

random from
top k by SfM

harder positives



hard and diverse training examples

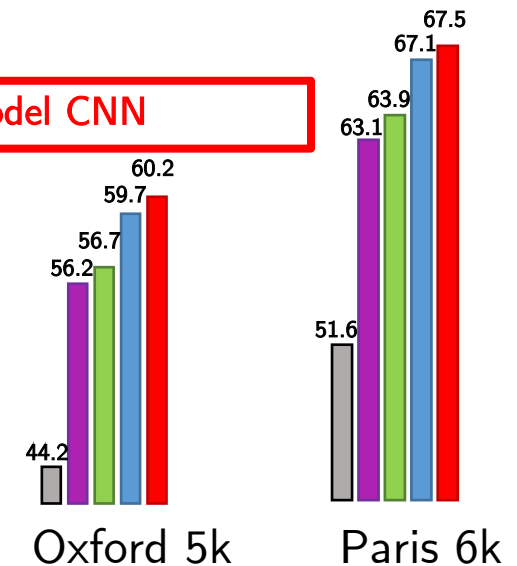
hard/diverse pos – hard/diverse neg: random(top k inliers) + top 1 / model CNN

hard pos – hard/diverse neg: top 1 inliers + top 1 / model CNN

easy pos – hard/diverse neg: top 1 CNN + top 1 / model CNN

easy pos – hard neg: top 1 CNN + top k CNN

no fine-tuning: pre-trained



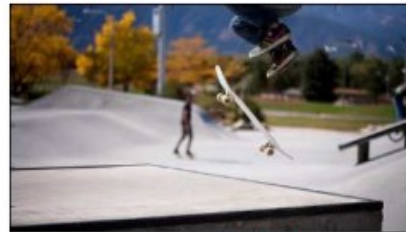
other tasks

semantic image retrieval

Training set (images + human captions)



- a boy **jumping** in the air with **black sneakers**
- a skateboarder doing a **trick**
- a skateboarder in the **air**
- **concrete** skateboard ramp
- a boy **lying** down



- A person is doing **trick** on the **skateboard**
- person **jumping** above **skateboard**
- The person is wearing **sneakers**
- a **concrete** skatepark
- railing on **ramp** surface
- a white **skateboard** in the **air**



- the person is **lying** on the ground
- the person is standing on the ground
- man holding a **frisbee**
- **shadow** of a person
- fingers on a **frisbee**

semantically similar

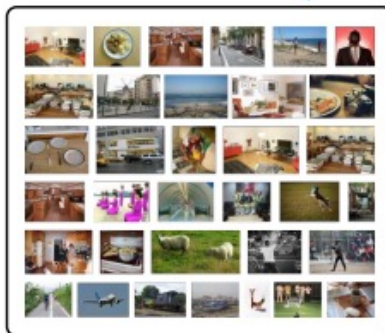
semantically dissimilar

At test time (images only)

Query image



Image database



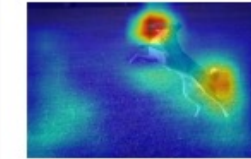
Semantic Retrieval



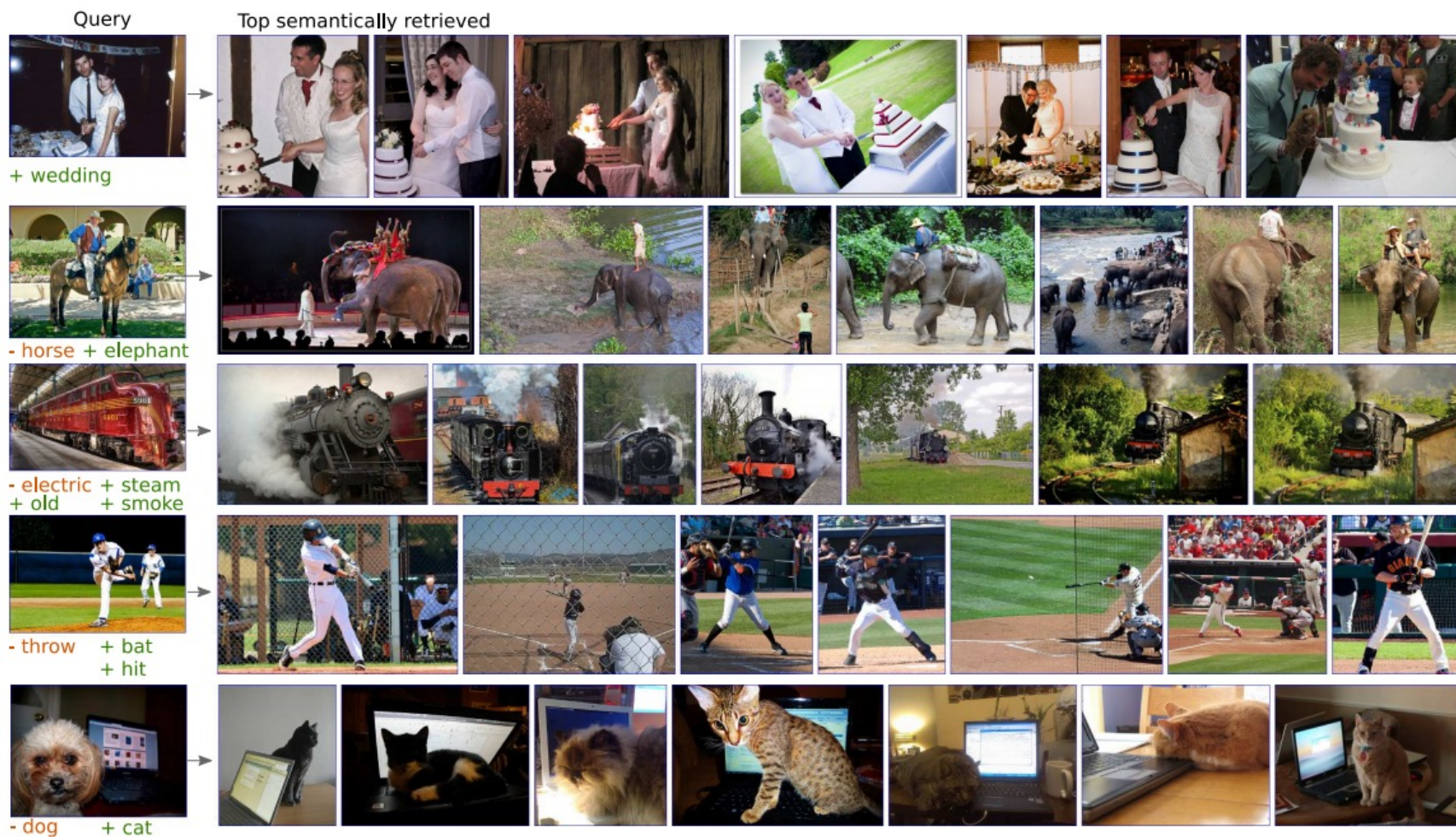
Retrieved image

Semantic representation

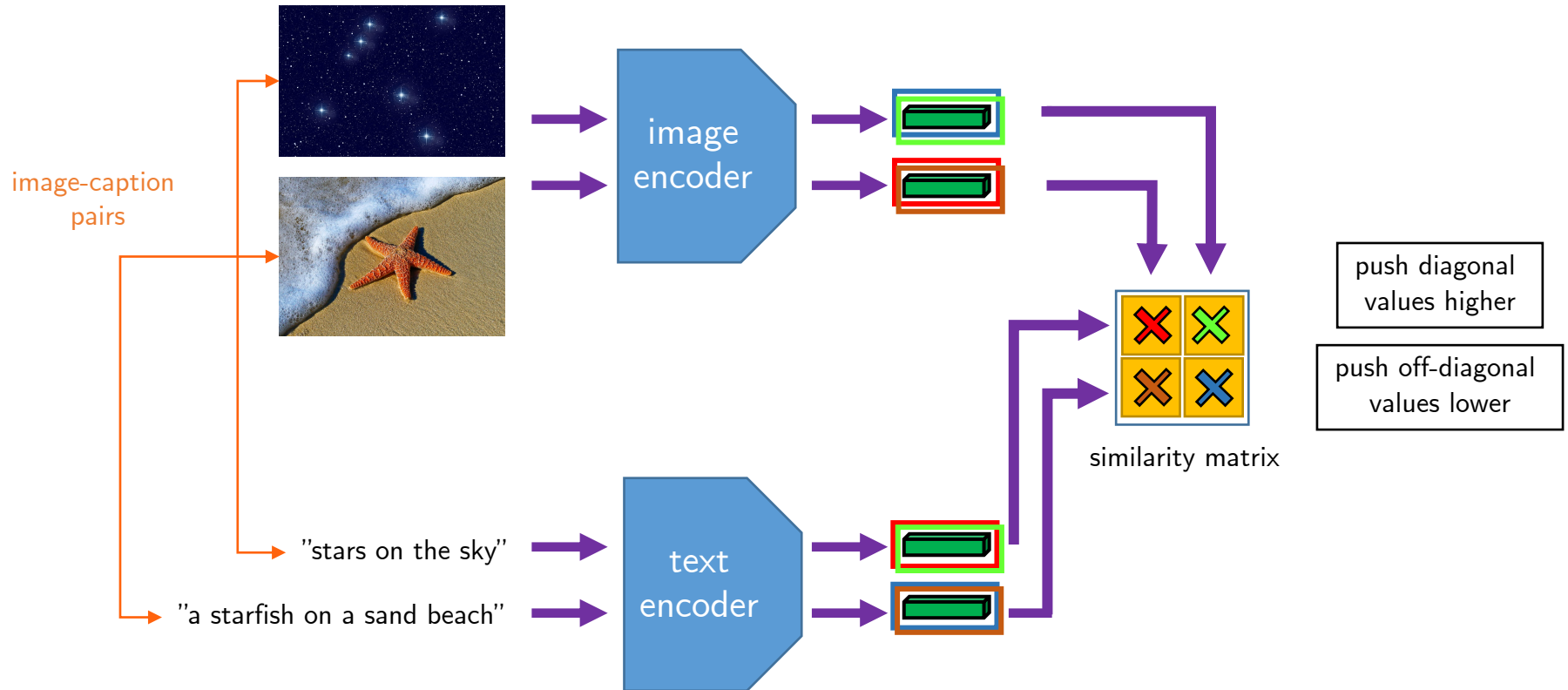
Grounding the decision



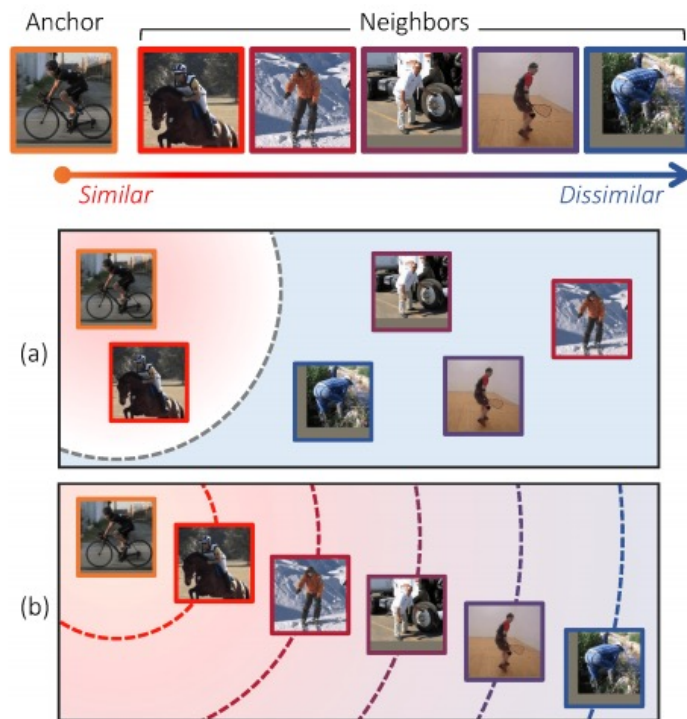
joint image-text query



CLIP – aligned text/image representation



beyond binary labels: retrieval for pose estimation



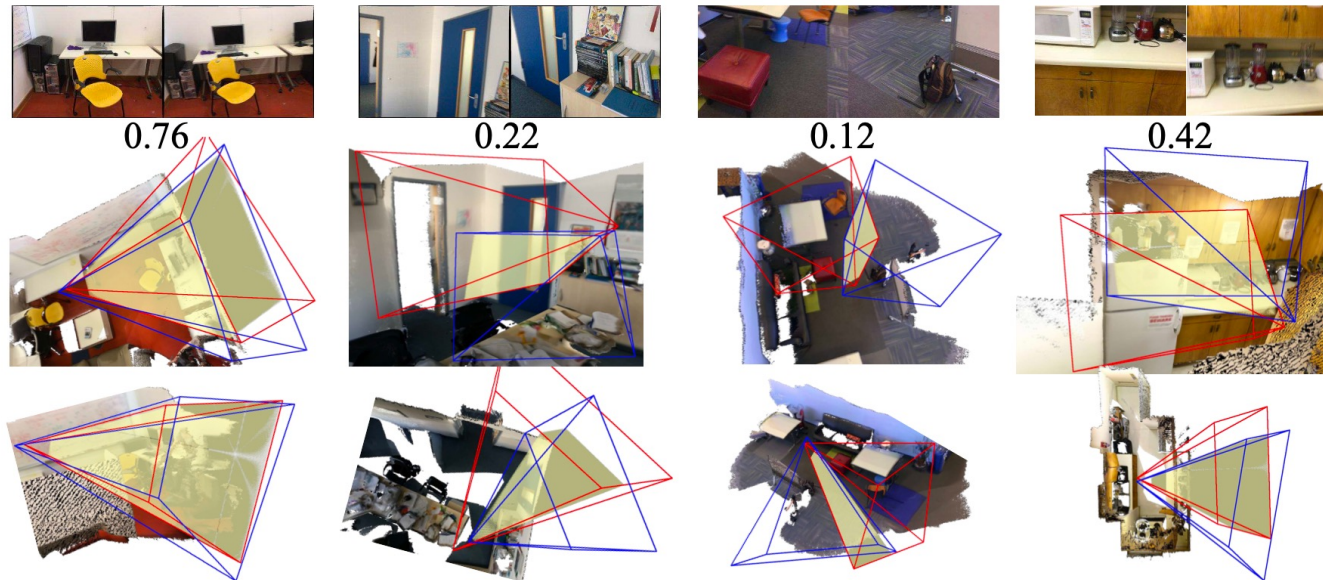
$$\ell(x_a, x_i, x_j, y_a, y_i, y_j) = \left(\log \frac{\|\mathbf{x}_a - \mathbf{x}_i\|_2}{\|\mathbf{x}_a - \mathbf{x}_j\|_2} - \log \frac{D(y_a, y_i)}{D(y_a, y_j)} \right)^2$$

↑
↑

distance ratio distance ratio
 representation space label space

camera overlap estimation

- pairwise label: camera overlap



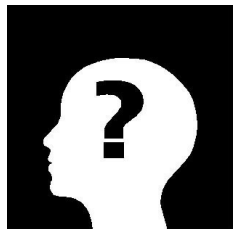
- train s.t. descriptor distance estimates camera distance

$$\ell(x_i, x_j, y) = (\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 - y)^2$$

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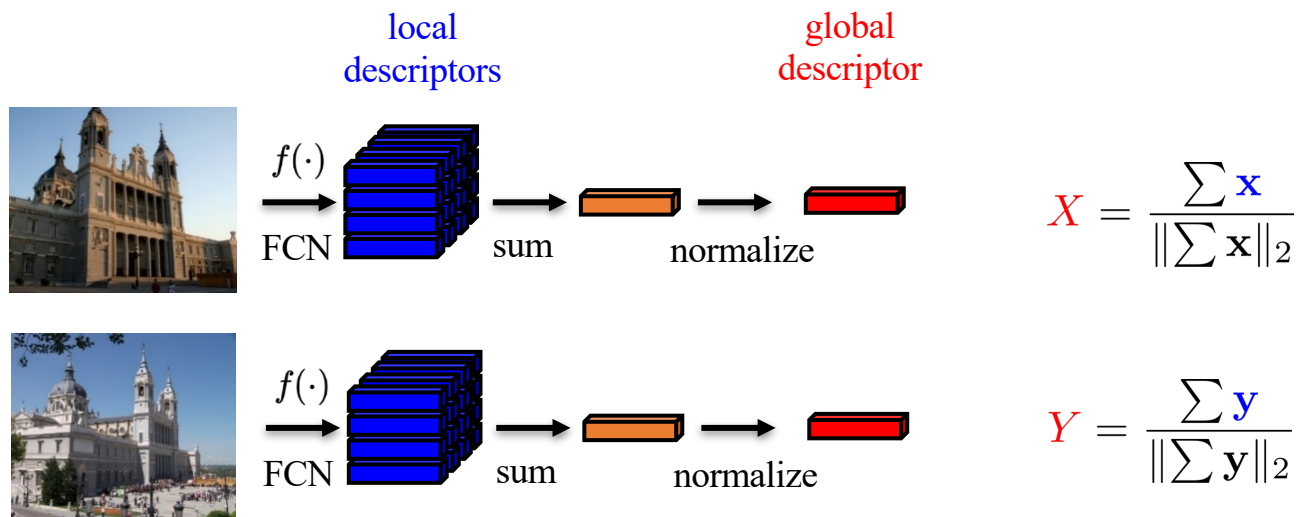
questions



metric learning of global
descriptors to learn local
descriptors

from global to local descriptors

- pairwise image labels
 - are weak labels for local descriptors
 - learn local descriptors without labels for image regions
- learning global descriptors generated with sum pooling (SPoC)
 - implicit learning of local descriptors



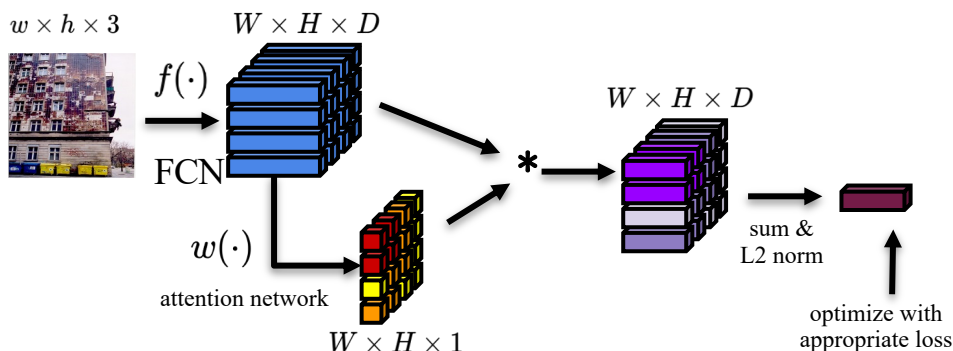
global descriptor similarity $\rightarrow$ $X^\top Y \propto \left(\sum \mathbf{x}\right)^\top \sum \mathbf{y}$

cross-matching $\rightarrow$ $= \sum \sum \mathbf{x}^\top \mathbf{y}$ $\leftarrow$ local descriptor similarity

deep local features - DELF

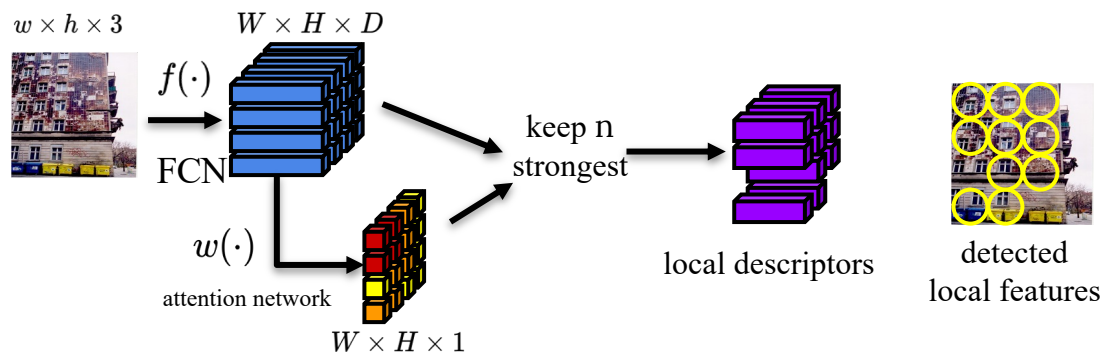
■ training

- weighted sum pooling: network-based weights
- use appropriate loss



■ inference

- keep local descriptors with largest attention weight
- reduce dimensionality with PCA
- index with ASMK or other approach



closer look

- global descriptor through weighted sum pooling

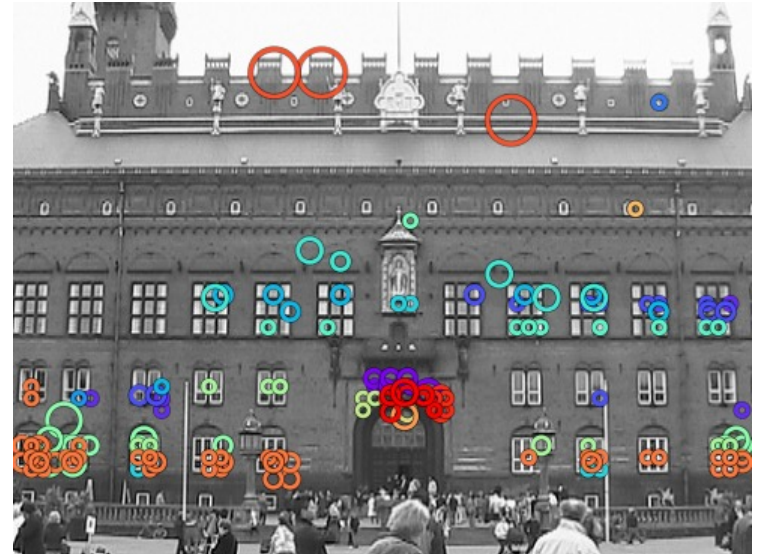
$$X \propto \sum_{\mathbf{x} \in \mathcal{X}} w_x \mathbf{x}$$

- image-to-image similarity interpreted as weighted cross-matching

$$X^\top Y \propto \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{y} \in \mathcal{Y}} w_x w_y \mathbf{x}^\top \mathbf{y}$$

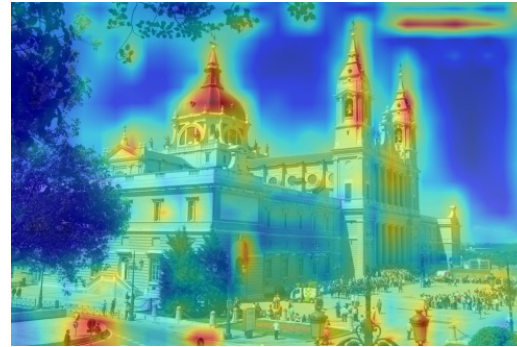
- background locations $\rightarrow$ small weights
- foreground locations $\rightarrow$ large weights
- matching locations $\rightarrow$ descriptors pushed closer
- non-matching locations $\rightarrow$ descriptors pushed apart

matching example using ASMK



20 visual words with the largest contribution are shown
different color per visual word

detection example



$w(\cdot)$ before our training



testing: detected features



$w(\cdot)$ after our training

no attention network
but use L2 norm
of activation vector
at each location instead

classical metric learning

metric learning: Mahalanobis distance

- learn a parametric distance function from the data
 - input examples are vectors
- example: Mahalanobis distance
 - M is a $D \times D$ positive semi-definite matrix
 - $d_M(\mathbf{x}, \mathbf{z}) = \sqrt{(\mathbf{x} - \mathbf{z})^\top M (\mathbf{x} - \mathbf{z})}$, $\mathbf{x}, \mathbf{z} \in \mathbb{R}^D$
 - $d_M(\mathbf{x}, \mathbf{z}) = \sqrt{(\mathbf{x} - \mathbf{z})^\top L^\top L (\mathbf{x} - \mathbf{z})} = \sqrt{(L(\mathbf{x} - \mathbf{z}))^\top L(\mathbf{x} - \mathbf{z})}$
 $= \sqrt{(L\mathbf{x} - L\mathbf{z})^\top (L\mathbf{x} - L\mathbf{z})} = \|L\mathbf{x} - L\mathbf{z}\|_2 = \|f(\mathbf{x}) - f(\mathbf{z})\|_2$
 - mapping function $f(\mathbf{x}) = L\mathbf{x}$
 - can be modeled by a single fully-connected layer

